

Impact of Listening to High-tempo Motivational Music During Cardio-respiratory Exercise Test on Treadmill

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ABSTRACT

Lack of motivation is a major factor which is associated with inhibition in physical activity and motivational music can play a significant role in this regard. 12 healthy and physically active subjects were involved in this study, who did 3 sessions of treadmill running exercise with a music intervention and another 3 sessions without any intervention. Pulse rates (PR), blood pressure (BP), rating of perceived exertion (RPE), feeling scale score (FS score) & Brunel music rating inventory-2 (BMRI-2) score was recorded in both conditions before, during and after exercise period and at early recovery period and data was analysed statistically. Result suggests that with musical intervention pulse rate significantly increased during exercise along with improvement of FS score and increase in post exercise systolic BP; but the other parameters did not alter significantly with musical intervention. So, from this study it is evident that, though PR and systolic BP were increased significantly during exercise in music condition compared to no music condition; with a simultaneous increase in feeling score (FS). This clearly demonstrates the positive reinforcement effects of the chosen high-tempo motivational music over cardio-respiratory exercise parameters. This type of music may not alter the feeling of exertion during intense exercise, but it can help in improving level of inclination for exercise and it has the capacity to provide positive effects for exercisers. Further research is needed to establish whether this type of music will have a similar beneficial effect on other forms of intense exercise in elite athletes.

Keywords: Music, Motivation, RPE, Pulse rate, Treadmill running

INTRODUCTION

Exercise is a form of planned, structured physical activity which should be an essential part of everyone's daily life. Lack of proper physical activity is one of the major and principal risk factors for

most of the non-communicable diseases including diabetes, obesity and cardiovascular diseases, which are the leading causes of death both in developed and developing countries (Radford et. al.,

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2018; Wanner et. al., 2017). It is also well-documented that a lack of motivation is often associated with an inhibition in physical activity (Williams, Rhodes & Conner, 2018). In other words, a high level of motivation before and during exercise makes the participants adherent to the task for a prolonged period through a psychophysiological stimulation. Music can play a significant role in this regard. Music is thought to affect human senses by allowing the separation of the thought from feeling, so that it alters one's perception of unpleasant feelings (Szmedra & Bacharach, 1998).

Humans have discovered the benefit of music on wellness since pre-historic time. Music is playing a noticeable role in sports and exercise activities since the ancient Greek culture, in which training procedures as well as events at the early Olympic Games were often practiced accompanied by music (Smirmaul, 2017). For past few years, music has been constantly gaining importance in the domain of exercise and sports science (Chtourou et. al., 2015; Feiss et. al., 2021). Music has been a fundamental aspect of human culture, which may even predate the verbal communication. Music has some of its own characteristics like tempo, quality of music (i.e., relaxing, motivational etc.). Listening to some high-tempo music while engaging in a physical activity is one of the common practices for many athletes and exercise participants. The scientific research in

this area has been prolific for the last few decades. This is further emphasized by the fact that more than one hundred peer-reviewed published articles have been found on PubMed, related to several aspects of the music-exercise correlationship.

Motivational music generally has a fast tempo (> 130 beats per minute [bpm]), a strong rhythm, inspiring lyrics and an uplifting harmonic structure, all of which collectively tend to increase vigour and induce bodily action. Although several coaches around the globe recommend use of motivational music during exercise and training, there is only a very small number of scientific studies has been done in India in this area. This is one of the major driving forces to take up this study involving such a relevant and comparatively less explored area of research. The specific aim of this present study is to examine the effect of high-tempo motivational music on exercising adults in terms of selected physiological and psychophysiological variables.

MATERIALS AND METHODS

All procedures related to this study were conducted in exercise physiology laboratory at Ramakrishna Mission Vidyalyaya Coimbatore campus. 12 healthy and active male participants of 25-35 years were selected and recruited in this study. Participants were involved in physical fitness and they had participated in State or National level of competitions

before and they had no prior exposure to similar type of study. Medical history & informed consent were obtained from each of the participants. Basic study related information like their age (30.42 ± 1.34), height (169 ± 1.58), and body mass (71.02 ± 1.26) was obtained before the experiment.

Parameters used in the study:

Pulse rate (PR): Pulse rate was measured during resting condition, at each 2-minute interval during exercise and at 2nd and 4th minute of recovery period. It was measured by using S-cure fingertip pulse oximeter (Model No. FTP-101).

Rate of perceived exertion (RPE): RPE of participants were measured at each 2-minute interval during exercise using Borg's CR10 RPE scale (Borg, 1998).

Blood pressure (BP): Blood pressures (both systolic and diastolic) were measured using a sphygmomanometer (DIAMOND Original Mercurial Regular BP Monitor). It was measured at resting condition & post-exercise condition (within 30 seconds of the termination of exercise) in each session.

Affective state: It was measured at each 2-minute interval during exercise in terms of feeling scale score using The Feeling Scale (FS) developed by Hardy and Rejeski (1989). This scale is an 11-point bipolar single-item scale that ranges from "very good" (+5 SCORE) to "very bad" (- 5 SCORE) along a pleasure-

displeasure continuum. FS score has been established previously as a valid and reliable measure of exercise-related affective states (Hardy & Rejeski, 1989). Motivational ratings of music used: The six-item Brunei Music Rating Inventory-2 (BMRI-2) was used in the study to measure the motivational components of music played during exercise (Karageorghis et. al., 2006). The BMRI-2 is a valid and reliable tool which has been used to select and standardize music played during experimental exercise protocols (Karageorghis et. al., 2006). Participants rated each item on a seven-point Likert scale ranging from "strongly disagree" (1 score) to "strongly agree" (7 score).

Procedural overview: Each of the participants made a total of seven visits to the laboratory over the course of approximately 3 weeks. At the first visit, participants were familiarized with the testing protocols & materials. In the next visits, they completed experimental trials in two conditions i.e., 3 sessions with music and 3 sessions without music. All sessions were performed on the same motorised treadmill (Motorised Treadmill TP-1090; TOPPRO). Treadmill was set up directly facing a wall in order to reduce the potential for external distractions. Steps followed in each session in described in figure 1.

The gaps between two experimental trials were 24 hours. In order to reduce diurnal variation in performance and

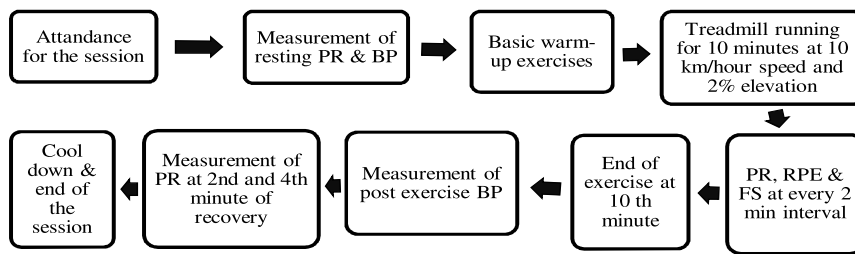


Figure 1: Steps followed in each session

other parameters, efforts were made to schedule participants at the same time of day for each visit to the laboratory.

A popular high-tempo motivational music (135 beats/min) was selected for this study liked by all the participants with almost an identical motivational score (BMRI-2). This music is usually being played in the gymnasiums and athletic training sessions. In the first day of experimental trial, BMRI-2 score (37.6+3.7 out of 42) was calculated from the participants' responses which gave the motivational ratings of that selected music. Based on similar type of study results, it was found that the motivational ratings were very satisfactory (36-42 score was considered to mark the music highly motivational in previous studies) (English, Mavros & Jay, 2019). So, the study was continued using that specific music. The music was played from in-built high-fidelity speakers in the treadmill at 75% of its maximum volume (it was well under auditory comfort level (50-60 decibels).

Statistical Techniques: All data were analyzed using Microsoft Excel 2016 and

its statistical plugin software Statplus LE (VERSION 7.3.0). Differences between no music and music condition in PR and BP in every time point were compared using paired t-test. RPE and FS scores in music and no music conditions were compared by using non-parametric statistical analysis (Mann-Whitney U test). In both tests, the level of significance was set at $p < 0.05$. Correlation coefficient 'r' was calculated in between 5 time period data of PR and feeling scores in 'with music' condition to see if any correlation exists in between them. Later, the exercisers' PRs were normalized into three time points: 1. Initial burst (0-4 min), 2. Middle steady phase (4-8 min) and 3. Final exhaustion phase (8-10 min).

A series of 2 Conditions [(no-music, music) $\times$ 3 (Time Point)] mixed-model one way analysis of variance (ANOVA) was conducted on pulse rates. Where a significant difference was observed, data were subjected to a Tukey's post hoc test to determine specific differences between conditions. Significance was set at $p < .001$. All values are expressed as means $\pm$ SEM.

RESULTS

Pulse Rate (PR): Primarily a small statistically insignificant ($p < 0.05$) difference was identified between conditions at resting baseline (Music: 73.7 ± 0.96 , No-music: 74.1 ± 0.99). With further analysis, a significant effect was found in experimental condition (with music) at all the time points during exercise when compared with controlled condition. The differences are significant in 2nd, 4th, 6th and 8th minute of PR, but not significant at the final minute of exercise. A small magnitude of difference was found in recovery PR (2 & 4 minutes of recovery). The pattern of PR change with progression of exercise and recovery in both music and no music condition is shown in figure 2.

To examine the pattern of increase in PR in music condition in comparison to

no music condition during exercise, the 5 time point data (2nd, 4th, 6th, 8th & 10th minute) was normalized to 3 time point data (initial burst of exercise, middle phase of exercise, final exhaustion phase of exercise) and percentage of increase in PR from no music condition to music condition was calculated. In ANOVA, it is seen that that percentages of increase in 3 time point were statistically differed ($p < 0.001$) in between them. It means that, the rate of increase in PR was not equal every time. From the mean values also, it is seen that during 2nd and 4th minute, the difference in between two condition is large. But with progression of exercised the difference in PR seems to be decreased and at 10th minute it becomes statistically insignificant. Figure 3 shows the percentage of increase in PR at 3 time points.

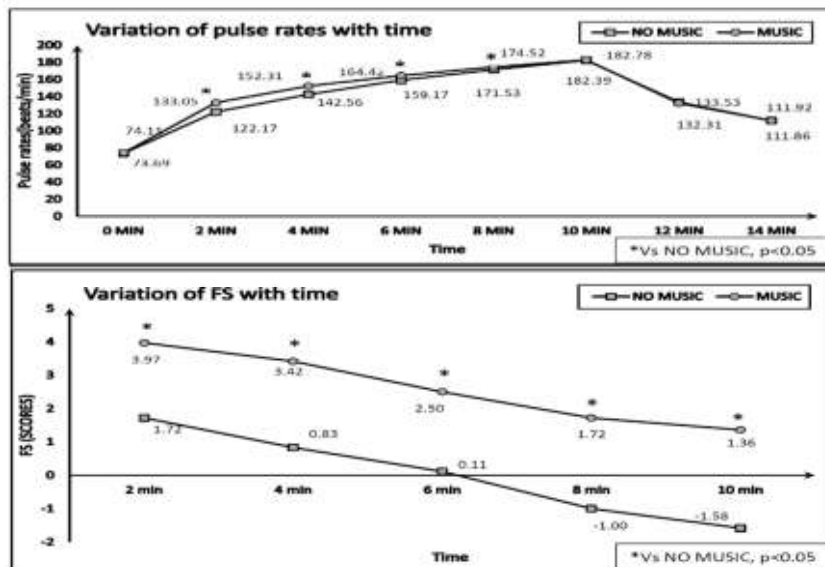


Figure 2: Variation of PR&FS with time

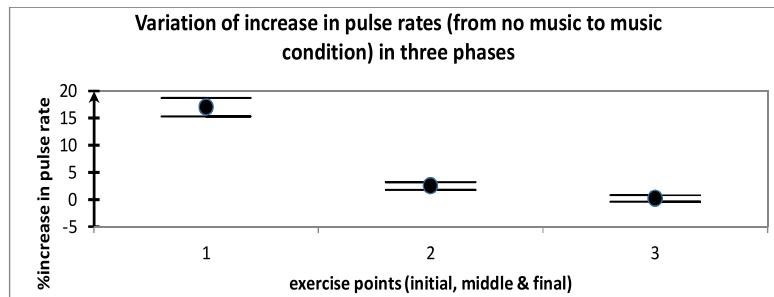


Figure 3: Increase in PR from no music to music condition at 3 time points

Rate of perceived exertion (RPE): At each of the 5 time points during exercise, the magnitude of RPE in both condition was similar in magnitude. There was no significant difference of RPE in between the conditions and so, no effect of music was found on RPE. The pattern of change of RPE with the progression of exercise in both experimental and controlled condition is shown in figure 4.

Blood pressure (BP) : Pre-exercise resting BPs (both systolic and diastolic) were similar in both conditions with a small and insignificant difference in magnitude. Then, post exercise systolic BP was significantly higher in music condition than in no music condition ($p < 0.05$). This is shown in figure 5.

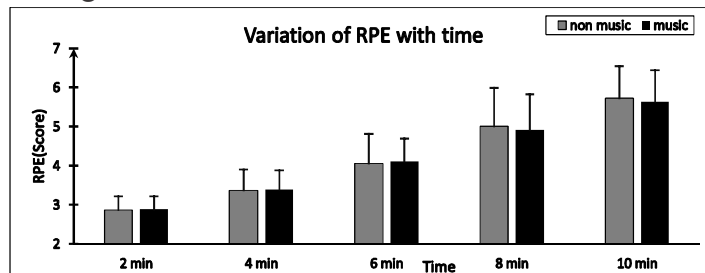


Figure 4: Comparison of RPE between two conditions at various time points

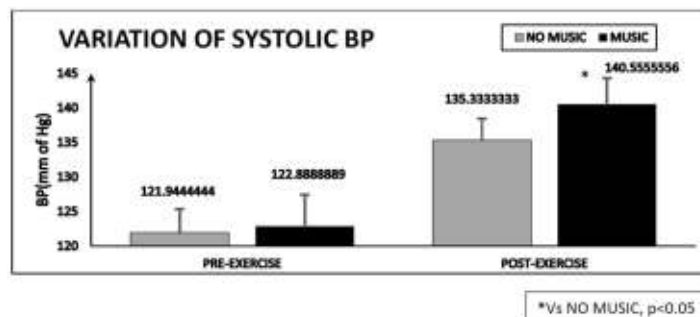


Figure 5: Comparison of pre & post-exercise BP between two conditions

Feeling scale score (FS score): At each of the 5 time points during exercise, feeling scale score differed in between two conditions. Participants in the music condition tended to report significantly

higher FS ratings ($p < 0.05$) across all time points during exercise which is shown in table 1 and the difference in between two conditions at every time period is shown in figure 2 and figure 6.

Time-point	Condition	
	non music	music
2 min	1.72 $\pm$ 0.07	3.97 $\pm$ 0.08***
4 min	0.83 $\pm$ 0.10	3.42 $\pm$ 0.08***
6 min	0.11 $\pm$ 0.16	2.5 $\pm$ 0.09***
8 min	-1 $\pm$ 0.16	1.72 $\pm$ 0.17***
10 min	1.58 $\pm$ 0.21	1.36 $\pm$ 0.17***

TABLE 1: FS scores at various time points of exercise (means $\pm$ SE.) (*Vs no music, $p < 0.05$)

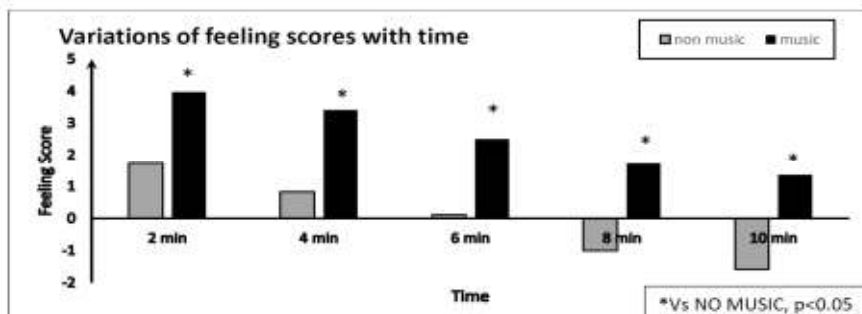


Figure 6: Comparison of FS between two conditions at various time points

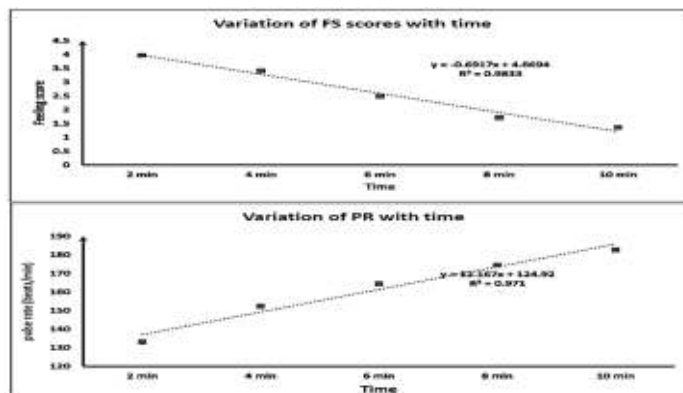


Figure 7: Comparison of correlation coefficient values of PR and FS in music condition

Relation between PR and FS scores: From the mean values it is seen that with progression of exercise, PR increases and FS scores decreases. But the pattern variation of PR & FS scores (in music condition) with time are found to be highly symmetrical as the correlation coefficient value for each of them was identical as in figure 7.

DISCUSSION

The main findings of the current study include-Exposure to high-tempo motivational music increased PR at all-time points during exercise, but it couldn't alter the exertion level of participants during exercise; it increased the post exercise systolic BP of participants significantly and resulted in an increase in affective responses in terms of feelings score during exercise than exercising at no music condition.

In various studies, high-tempo music was used during exercise which increased the HR significantly than in no music condition (Feiss et. al., 2021). Similarly, some studies used motivational music or arousing music or motivational music and compared its effect with the effect of relaxing music during exercise and reported that arousing music increased HR at various time points during exercise and recovery (Elliott, Polman & Taylor, 2014; Kuan et. al., 2018). But there was a good number of investigations also, in which researchers found a decrease in HR

or no change. This is mainly due to differences in duration and intensity of exercise and type of music used (Szmedra & Bacharach, 1998; Centala et. al., 2020). In this study, no significant difference was observed for recovery PR which was similar to Michal Eliakim et al as they had also used motivational music in their study (Eliakim et. al., 2012).

Musical perception involves conscious cortical as well as unconscious sub cortical regions of brain and the responses associated with listening to music is very complex (Zatorre, Chen & Penhune, 2007; Levitin & Tirovolas, 2009). The rhythmic pattern of some music can actually help in movement execution (Levitin & Tirovolas, 2009). According to P. N. Juslin's theory of bio-musicological process of rhythmic entrainment (Juslin, 2013), the rate of movement and bodily pulse such as heart rate and respiration rate are drawn toward synchronization with the rhythmical qualities of music. This explains why listening to a high-tempo music causes an increase in heart rate. It can also be explained by the brain stem reflex caused by listening to motivational music, by which central nervous system becomes activated and increases the HR (Juslin, 2013). So, along with brain, music also influences the whole body and this occurs through physiological arousal mediated by sub-cortical structures and bodily rhythms such as movement, breathing rate or heart rate. People prefer a high

tempo in intense exercise for this reason. Previous studies also suggest that, music can influence autonomic nervous system and so, it can be used to regulate hemodynamic variables like BP and HR (Harmon, & Kravitz, 2007, Murrock & Higgins, 2009). Increase in HR was found to be more in initial burst of exercise than in middle steady phase or final exhaustion phase which suggest that motivational music is able to increase cardiovascular efficiency at an early stage of exercise.

In this domain the findings for RPE are almost similar with most of the previous studies (Feiss et. al., 2021; Maddigan et. al., 2019; Edworthy & Waring, 2006; Centala et. al., 2020; Dyer & Mckune, 2013; Eliakim et. al., 2012; Stork et. al., 2015) with some exceptions. Some of them included endurance exercise while some other used resistance exercise. So, it is evident that pattern and type of exercise didn't influence the findings related to effect of music on perceived exertion. Few of the previous researches reported some changes in RPE while exercising with music (Szmedra & Bacharach, 1998; Patania et. al., 2020; Nikol et. al., 2018). This was probably due to difference in choosing the music. Some of the previous researches reported an increase in perceived exertion level with the progression of time (Jones, Tiller & Karageorghis, 2017.; Lopes-Silva et. al., 2015), which supports our findings. It is mostly due to greater accumulation of

metabolic factors generated due to exercise, which causes fatigue and increases the feeling of exertion.

Previous studies indicated that strong sensory stimuli like auditory stimuli elicited by music may inhibit physiological feedback signals associated with feelings of fatigue (Rejeski, 1985). Experimental work using EEG has shown the effectiveness of music in reducing theta waves (4–7 Hz) in the various regions of the brain like frontal, central, parietal, and occipital regions (Bigliassi et. al., 2016). Suppression of fatigue related symptoms is thought to be directly associated with this process (Craig et. al., 2012). But, in this study, physiological feedback signals predominated over psychological system of the participants, which was reflected in the study evident by no change in RPE in experimental condition.

All researches related to effect of music tempo on BP reported that use of high tempo music increases BP and use of slow tempo relaxing music reduces BP (Savitha, Mallikarjuna & Rao, 2010; Birnbaum, Boone & Huschle, 2009; Knight & Rickard, 2001). All these studies support our findings. This rise in BP can be described similarly using brain-stem reflex phenomenon. Also, the pleasure factor contributed to this by increasing the systemic blood flow as evident by Miller et al (2008) study (Miller et. al., 2008)

This study indicates that affective states of participants in terms of feeling scale score (FS), which was found to be much higher in music condition compared to no music condition. Almost all past studies have reported similar findings (Elliott, Polman & Taylor, 2014; Belkhir et. al., 2019). It was also reported that fast tempo music gives better FS score than slow tempo music (Jones, Tiller & Karageorghis, 2017.). A few studies have reported that they found no significant difference in FS score in between conditions but they found significant changes of feeling score with time (Stork et. al., 2015). But in our study, both in between conditions and in between several time points, differences were found.

In this study, it was found that the pattern of change in PR and FS scores with time in music condition were almost in similar fashion and correlation coefficient 'r' for both has identical values. It signifies that change in PR and affective state of a participant during exercise in music condition is highly correlated. So, from this study it is evident that, though HR and systolic BP were increased significantly during exercise in music condition compared to no music condition; with a simultaneous increase in feeling score (FS). This clearly demonstrates the positive reinforcement effects of the chosen high-tempo motivational music over cardio-

respiratory exercise parameters in this study.

CONCLUSION

The current findings indicate that motivational music may not alter the feeling of exertion during intense exercise, but it can help in improving level of inclination for exercise. This finding is of relevance to practitioners and coaches working with athletes as well as recreational exercisers, as it offers an easily implementable intervention by which to increase the motivation level during a highly physically demanding session. This type of music is also physiologically beneficial as it increases PR during initial phase of exercise for a better blood flow shift which facilitates the body to shift its physiological systems in their higher activated state from stationary phase to exercise. These results support the notion that music can modify the interplay between central motor drive, central cardiovascular command, and affective responses. So, it is reasonable to conclude that high-tempo music has the capacity to provide positive effects for exercisers. Further research is needed to establish whether this type of music will have a similar beneficial effect on other forms of intense exercise in elite athletes. Also, other types of music may be tested in a similar set up to reach a conclusion regarding the tempo and quality of the tunes related to motivation and exercise efficacy.

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